Key Visual: The Traveler's ivory-white echo pulse colliding with the Siren's magenta interference across the Glacial Ocean.

An Echo Has No Shape

Unity 6 (URP) · C# · 4-Week Solo Dev

High Concept & Design Premise: A first-person sightless acoustic labyrinth built inside a continuous `550m × 550m` fog basin for Peter Brinson's course at USC (*Situated Player Role: The Traveler — World as Presence*). Rather than adding ability creep, combat, or non-diegetic UI, the game locks the player to a **single unchanging verb (Press `E` to Cast Echo)** and drives progression by subverting the epistemic rules of how each region answers back.

550m × 550m	1-Button Verb (E)	4-Tier Wayfinding	Telemetry Tuned
Seamless Single-Scene Basin	Zero Ability Creep · 3 Shifts	Zero HUD · Zero Minimaps	90% Silence Discovery Rate

1. Core Design Philosophy: The Traveler, Presence, and One-Verb Restraint

USC CTIN · Situated Role: The Traveler (World as Presence)

1.1 Origin: World as Presence

Presence

In Peter Brinson's taxonomy, the **Traveler** differs from the Investigator in how meaning is constructed: rather than piecing together forensic clues about a past event, the Traveler experiences a world in the present tense through movement, attention, and tolerance of absence. Where the player walks, when they pause, and what they stop to hear determine what the world reveals—moving from *feeling lost in wonder → recognizing a pattern → finding their way through interpretation.*

1.2 One Verb in One Month

Single Verb

Most echolocation games treat sound as a binary visibility flashlight: once the pulse reveals the room, the mechanic stops evolving because the relationship between input and returned information remains static. Keeping the player's action set fixed to a single button (`Press E to Cast Echo`, plus `F` to Attune) solved both a philosophical problem and a 4-week scoping problem—focusing the entire month on destabilizing how the world answers back.

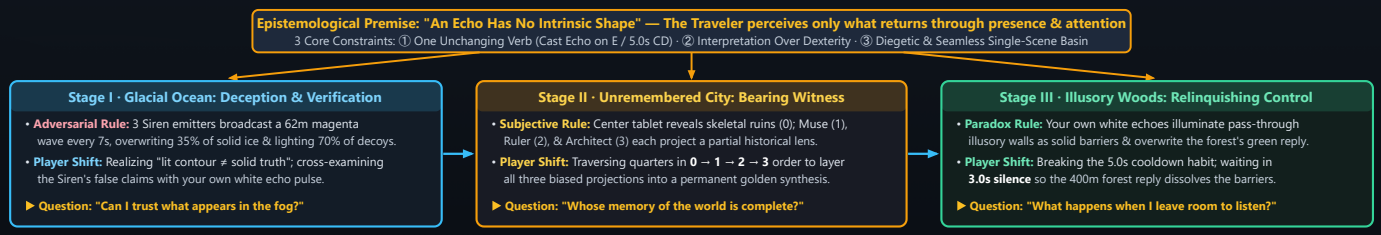
1.3 The Epistemological Core

Epistemology

The title is literal: **an echo has no intrinsic shape of its own**. It is a formless wave that borrows the contour of whatever surface stands in its way, and what returns is always partial, delayed, or distorted. Across the three regions, the Traveler's relationship to their own voice shifts from **skeptical verification** (Glacial Ocean), to **bearing witness** (Unremembered City), to **relinquishing control** (Illusory Woods).

Fig 1.1 | Epistemological Premise, Systemic Constraints & The Traveler's 3-Stage Experiential Arc

Three Epistemic Inversions Under a Single Unchanging Input (`Press E`)



2. The "Rule of Three" Epistemic Design Matrix

Subverting Default Player Assumptions Across Three Regions

Design Dimension	Hub: Stone Monument	Region I: Glacial Ocean	Region II: Unremembered City	Region III: Illusory Woods
Epistemological Stage	`Calibration` Baseline sonar & hub orientation	`Verification` Cross-examining adversarial noise	`Synthesis` Layering biased historical lenses	`Relinquishing Control` Pausing in silence to hear the reply
Epistemological Question	*"How do I give shape to silence?"*	*"Which signal in the fog is lying?"*	*"Whose memory of the past is complete?"*	*"What happens when I leave room to listen?"*
Player Habit Subverted	Relying on permanent sight and UI waypoints.	Treating every lit contour in the mist as solid geometry.	Assuming a single observer's projection holds the whole truth.	Casting echoes automatically on cooldown (every `5.0s`).
Acoustic Counterpart	**The Prophecy Tablet** (Static Central Anchor)	**The Siren Emitters (×3)** (`62m` Magenta Deception Pulse)	**Muse, Ruler & Architect** (`150m–200m` Perspective Waves)	**The Forest Reply Center** (`3.0s` Delayed Call & Response)
Source of Friction	Initial disorientation in open volumetric mist.	Distinguishing solid stepping stones from non-solid decoys.	Out-of-order attunement or exiting bounds triggers a `45m/s` reset.	Speaking too soon overwrites the green reply & snaps barriers shut.
Chromatic & Diegetic Hint	Obsidian Teal & Pale Sage *"Three silences wait in the mist..."*	Glacial Slate & Siren Magenta *"Not every voice across the ice wishes you to reach the shore."*	Sky Blue / Crimson / Emerald → Gold *"The city stands whole within the meeting of many eyes."*	Deep Pine & Verdant Green *"The woods have opened to the voice you left room to hear."*

Fig 2.1 | Master World Map (~550m × 550m): Single-scene layout (`Main.unity`) showing the central Stone Monument Hub, nested inner and outer regional fog bounds, outskirts tablets, and connecting acoustic guide splines.

3.1 Single-Scene Hub-and-Spoke Topology

`Zero Loading Screens`

To maintain **zero HUD markers, minimaps, or scene transitions**, the entire `550m × 550m` acoustic basin lives inside one continuous Unity scene around the central **Stone Monument Hub**. Players awaken at the monument, attune to the prophecy tablet to ignite initial guide splines, and explore the three regional labyrinths in **any non-linear order**:

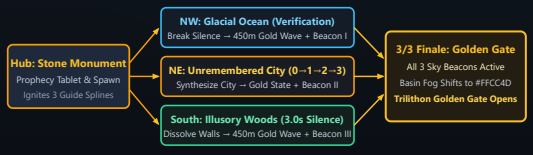
- **NW Basin · Glacial Ocean:** Descends from the western shore across branching ice floes toward the northern terminal tablet and sky beacon.
- **NE Plateau · Unremembered City:** Crosses the West Bridge causeway into a `200m × 220m` street grid enclosed by an outer reset boundary.
- **Southern Valley · Illusory Woods:** Accessible from both the NW hub spline (Tablet `T1`) and SE city approach (Tablet `T2`), leading to the SW Sentinel Tower.

3.2 Dual-Collider Atmospheric & Audio Crossfades

`Nested Bounds`

Without a visible horizon line, regional transitions rely on **diegetic atmospheric and musical crossfades**. Each region is wrapped in nested `OuterBounds` and `InnerBounds` trigger volumes. As the player walks across the outskirts buffer (normalized distance `t ∈ [0, 1]`), the environment controller smoothly interpolates **volumetric fog color, fog attenuation distances (D_{start}, D_{end}), and regional music stems**.

Fig 2.2 | Non-Linear Hub Flow & 3/3 Golden Gate Finale Global Progression Aggregator



3.3 Four-Scale Sightless & Diegetic Wayfinding Stack

`Orienting Across 550m Without HUD Arrows or Minimaps`

Wayfinding Scale	Effective Range	Diegetic Mechanism	System & Shader Implementation	Navigational & Experiential Function
1. Macro Scale `Basin-Wide`	`150m – 450m`	**World Reverberations & Sky Beacons**	Breaking a region's silence triggers a `450m` golden wave (`12s` hold) and ignites a vertical sky beacon (`BeaconGlacier/City/Woods`).	Provides basin-wide macro-orientation across the fog, rewarding regional completion and drawing the player back toward the Hub or unvisited regions.
2. Meso Scale `Regional Buffer`	`50m – 150m`	**Dual-Collider Fog & Music Crossfades**	Interpolates volumetric fog color, visibility falloff, and regional music stems as the player crosses between `OuterBounds` and `InnerBounds` .	Communicates regional threshold crossings through shifting color temperature and musical texture before specific architecture becomes visible.
3. Micro Scale `Arterial Paths`	`15m – 50m`	**Acoustic Guide Splines** (Shimmering Threads)	`SplineContainer` ribbons along arterial routes that remain dormant until struck by an echo, then send shimmering particles along the path tangent.	Preserves active discovery: arterial routes answer back only when the player calls out, guiding transitions without persistent floor arrows.
4. Immediate Scale `Local Space`	`0m – 15m`	**Seen vs. Unseen Visibility Hierarchy**	Separates assets into `Seen` (faint permanent silhouettes in mist) and `Unseen` (`0%` base visibility, revealed only by spherical wave intersection).	Ensures primary anchors (tablets, twin ice spires, witness statues) remain readable between cooldowns while keeping secondary terrain hidden until echoed.

3.4 Central Hub Progression: Stone Monument (Initial State vs. 3/3 Golden Finale)

`0/3 Initial Awakening vs. 3/3 World Reverberation`

Fig 2.3 | Stone Monument Hub (Initial State 0/3): A player-cast white echo sweeps across the monolith ring and central prophecy tablet in the pale mist.

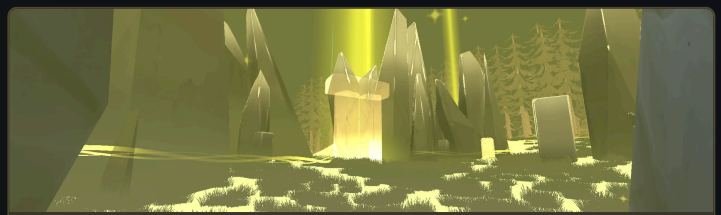


Fig 2.4 | Stone Monument Hub (3/3 Completion): Breaking all three regional silences unlocks the golden trilithon gate and illuminates the basin's guide splines and sky beacons.

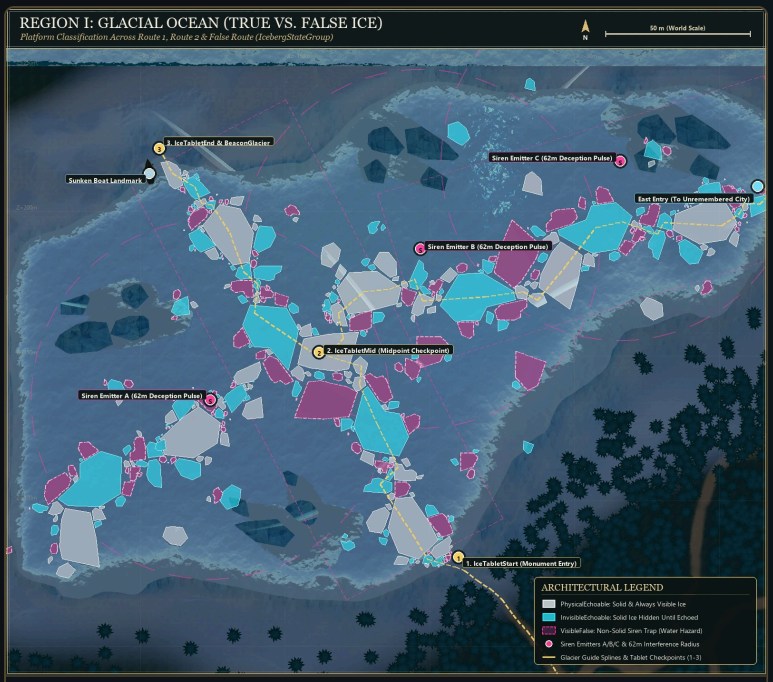


Fig 3.1 | Region I Level Design Map (Glacial Ocean): Showing solid anchor floes (white), hidden solid stepping stones (cyan), non-solid Siren traps (magenta dashed), and three `62m` Siren emitter radii.

The Design Problem: Evaluating Where to Jump

`Verification`

First-person platforming across ice floes becomes routine if the player's echo is completely reliable, and frustrating if jumps demand tight mechanical precision. The Glacial Ocean places the tension in **evaluating which signal to trust** rather than in dexterity. Three unseen **Siren Emitters (A/B/C)** broadcast a periodic `62m` magenta wave every `7.0s ± 1.0s` that corrupts nearby ice visibility and lures hasty players onto non-solid water traps.

Three-Tier Iceberg Visibility & Collision Hierarchy

`True vs. False Ice`

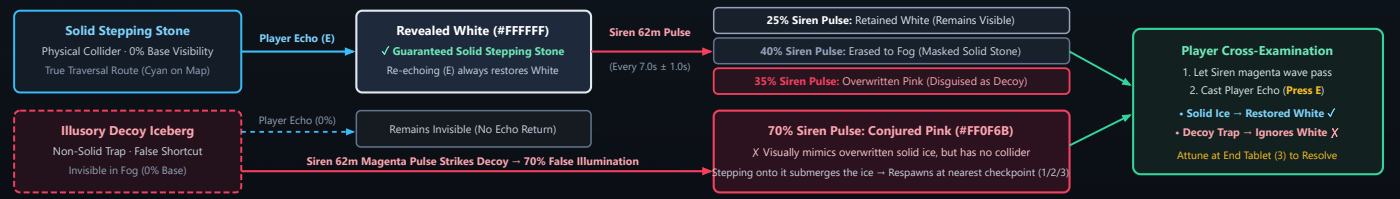
- **1. Solid Anchor Islands (White on Map):** Permanent islands and tablet checkpoints (`1` Start, `2` Twin Spires Midpoint, `3` End) that remain faintly visible in the mist as safe observation anchors.
- **2. Hidden Solid Stepping Stones (Cyan on Map):** Physically solid floes with `0%` base visibility. **Only the player's white echo reliably reveals them in pure white**; when struck by a Siren wave, **40% are erased back to fog** and **35% are overwritten in pink**.
- **3. Illusory Decoy Floes (Magenta Dashed on Map):** Non-solid traps with no collider (submerging on contact). **The player's white echo passes straight through them (0% return)**, but the Siren's pulse falsely illuminates **70% of decoys in bright pink**.

Spatial Layout & Cross-Examination Loop

`Shortcut Temptation`

Between Checkpoints `1 → 2 → 3`, non-solid decoy floes are deliberately arranged along tempting direct shortcuts, while true solid stepping stones curve along flanking arcs. Because a **Pink** iceberg could be either a `35%` overwritten solid floe or a `70%` non-solid decoy, players must wait for the Siren's wave to pass and cast their own `E` echo to cross-examine the ice before jumping.

Fig 3.2 | Glacial Ocean: Probabilistic Iceberg Visibility State Machine & Cross-Examination Loop — Player White Echo (100% Honest Ground Truth) vs. Siren 62m Magenta Pulse (Probabilistic Corruption & Decoy Conjuration)



4.1.3 In-Game Execution: White Echo Verification vs. Golden Route Resolution

`Cross-Examining Stepping Stones at the Twin Spires`

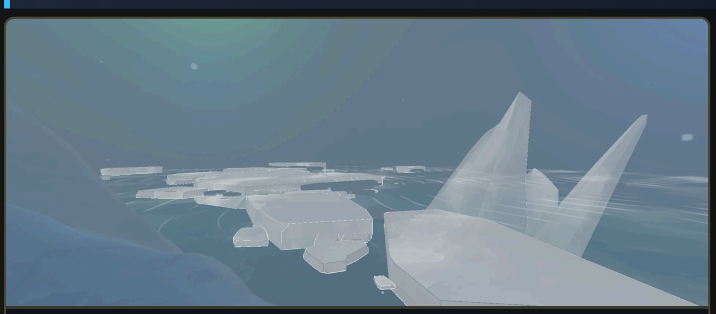


Fig 3.3 | White Echo Verification: A player-cast white echo revealing guaranteed solid stepping stones in front of the twin ice spires checkpoint.

Fig 3.4 | Golden Route Resolution: Attuning at the northern end tablet silences the Sirens and illuminates the completed Glacial Ocean route in gold alongside the guide spline ribbon.

Fig 4.1 | Region II Level Design Map (Unremembered City): Showing the central plaza tablet (`0`) and the three witness quarters—The Muse (`1` , SW), The Ruler (`2` , SE), and The Architect (`3` , NE)—plus the outer reset boundary.

The Design Problem: Spatial Storytelling Without Dialogue

Synthesis

How do you express the idea that **history is a collection of biased, incomplete projections over a ruin** purely through spatial layout and environmental state, with zero dialogue trees? No single observer's memory is sufficient to restore the city; only by traversing all three quarters in sequence and layering their architecture together does the city become whole.

Five-Stage Perspective Wave & Architecture Stack (Stage 0 → 4)

State Stack

- **Stage 0 · Central Plaza Tablet (Neutral Wave `#E6F2EE`):** Emits a `200m` neutral pulse that reveals the baseline street grid and skeletal ruins while leaving courtyards, gates, and structural spans absent.
- **Stage 1 · SW Quarter — The Muse (Sky-Blue Wave `#73CCFF`):** Remembers the city as a sanctuary of art and nature; shifts fog to cerulean and materializes **courtyard structures, birds, and living flora**.
- **Stage 2 · SE Quarter — The Ruler (Crimson Wave `#FF290A`):** Remembers the city as an instrument of authority; shifts fog to crimson and layers **monumental gates, colonnades, and citizen silhouettes** over the flora.
- **Stage 3 · NE Quarter — The Architect (Emerald Wave `#33FF61`):** Remembers the city as load-bearing geometry; adds **structural walls, arches, and buttresses**, culminating in **Stage 4 Golden Synthesis (`#FFCC4D`)**.

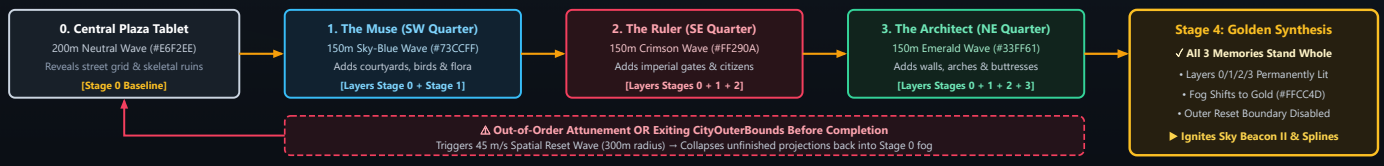
Sequence Verification & 45m/s Spatial Reset Wave

Collapse Rule

Attuning to a witness statue out of sequence (anything other than `0 → 1 → 2 → 3`) or stepping outside `CityOuterBounds` before completing the synthesis triggers a `45m/s` **Spatial Reset Wave** from the central plaza that sweeps across the plateau and collapses unfinished projections back into the fog.

Fig 4.2 | Unremembered City: 0 → 1 → 2 → 3 Cumulative Memory Synthesis State Machine & 45m/s Collapse Loop

Each Valid Attunement Preserves Prior Architectural Layers Toward Stage 4 Golden Synthesis



4.2.3 Unremembered City: Four-Stage Visual Synthesis Progression

Neutral, Sky-Blue, Crimson & Golden Synthesis States

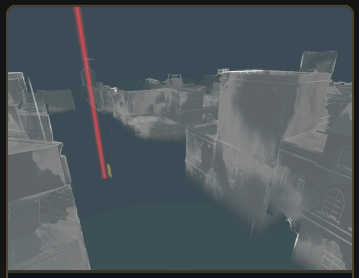


Fig 4.3 | Stage 0 (Neutral): Baseline skeletal ruins and the central plaza beacon.

Fig 4.4 | Stage 1 (The Muse): Sky-blue fog (`#73CCFF`), sanctuary courtyards, birds, and flora.

Fig 4.5 | Stage 2 (The Ruler): Imperial crimson fog (`#FF290A`), archway, and citizen silhouettes.

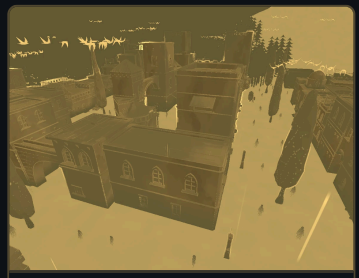


Fig 4.6 | Stage 4 (Golden Synthesis): All three perspective groups restored simultaneously in gold.

Fig 5.1 | Region III Level Design Map (Illusory Woods): Pass-through illusory barriers (bright green bars) block arterial junctions until dissolved by the `400m` forest reply wave after `3.0s` of silence.

The Timing Paradox: Your Own Echoes Sustain the Barrier

`Call & Response`

Inside the dirt-corridor labyrinth, 8 **pass-through illusory barriers** (`FalseWalls`) sit across the primary junctions leading toward the southwest clearing and Sentinel Tower. In the forest, **the player's own echoes maintain the barrier**: casting a white echo (`42m` range, `1.4s` duration) illuminates these illusory tree walls alongside real forest walls, making direct corridors appear blocked.

Systemic Solution: When the player casts an echo and then **waits in silence for** `3.0s` , the center of the forest (`WoodsEchoSource`) answers back with a `400m` **Verdant Reply Wave (** `#47FF6B` **at** `50m/s` **)** that dissolves all illusory barriers for `5.0s` . Conversely, pressing `E` during the hold window triggers a `0.35s` **Active Wave Overwrite** that carves through the green wave and snaps the barriers shut.

Fig 5.2 | Acoustic Guide Ribbon: Pine corridor illuminated by an echo-triggered guide spline.

Fig 5.3 | Sentinel Tower Clearing: Revealed in emerald glow after passing through dissolved barriers.

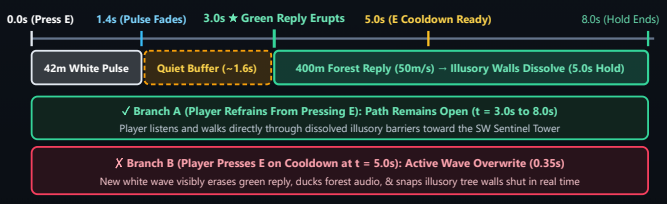
4.3.2 Playtest Telemetry: 3.0s Silence Window

`6-Cohort Data`

Reply Delay	Accidental	Intentional	Observed Playtest Behavior
`1.0s`	**85%**	15%	Players trigger the green reply while walking and echoing normally without noticing the silence rule.
`2.0s`	45%	50%	Mixed recognition; cautious players notice, rapid walkers trigger by accident.
`3.0s` ★	**10%**	**90%**	**[Tuned Sweet Spot]** Player pulse fades out (`42m`); leaves a clear `2.0s` quiet buffer before the `5.0s` cooldown finishes.
`4.0s`	5%	60%	Impatient players begin spam-pressing `E` near the end of their cooldown.
`5.0s` (CD)	0%	20%	Matches cooldown: players press `E` on cooldown, perpetually resetting the silence timer.
`6.5s`	0%	**5%**	Too long: players assume nothing happened and keep echoing in a loop.

Fig 5.4 | The 8-Second Call-and-Response Cadence & 0.35s Overwrite

Systemic Cause & Effect



5. Design Retrospective & Key Takeaways

`Mechanical Restraint · Diegetic Feedback · Spatial Alignment`

1. Mechanical Restraint

`Single Verb`

Keeping the player on a single, unchanging `E` input throughout the game meant that progression had to come from **recontextualizing how information is read rather than expanding the action set**. The input at the end of the game is identical to the input at the start, shifting from **verification** (Glacial Ocean), to **synthesis** (Unremembered City), to **restraint** (Illusory Woods).

2. Diegetic Feedback Over Text

`Cause & Effect`

In the Illusory Woods, diegetic hints on the outskirts tablets were not sufficient on their own to break the habit of echoing on cooldown. Tuning the delay to `3.0s` and introducing the `0.35s` **active wave overwrite** (where a new player pulse visibly erases the forest's green reply and snaps the walls shut) let players infer the silence rule purely from systemic cause and effect (**15%** → **90%** discovery).

3. Aligning Topology With Rules

`Level Layout`

Each region's physical layout reinforces its acoustic rule: branching ice routes where **non-solid decoy floes mimic convenient shortcuts** in the Glacial Ocean, **three witnesses separated across distinct quarters** of the Unremembered City so the player traverses the street grid to layer their perspectives, and **pass-through barriers placed across arterial junctions** in the Illusory Woods.